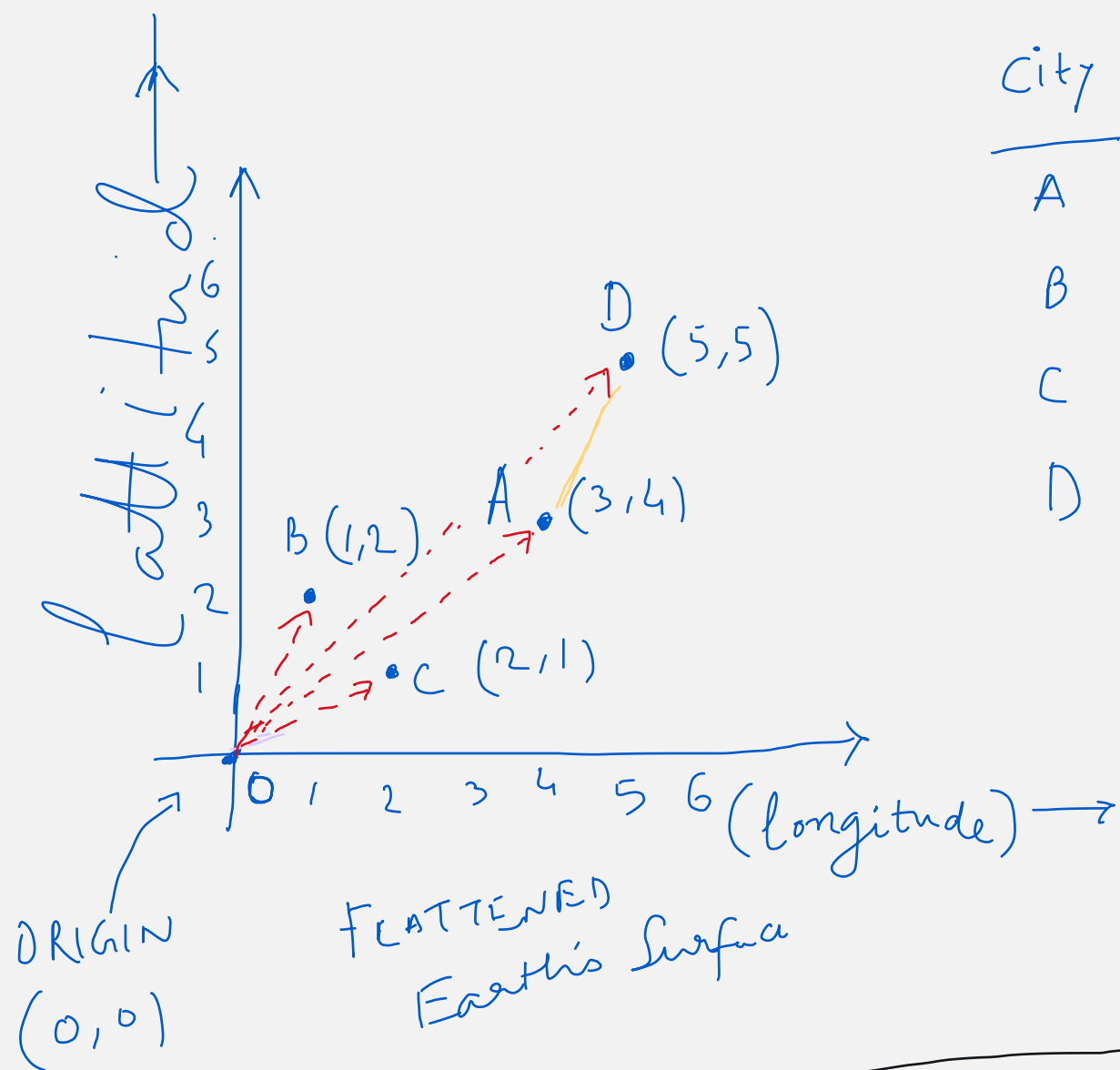




City	Longi.	Latti	vector
A	3	4	(3,4)
B	1	2	(1,2)
C	2	1	(2,1)
D	5	5	(5,5)

$$\begin{aligned}
 \text{Norm}(A) &= \text{distance of } A \text{ from the origin} \\
 &= \sqrt{(3-0)^2 + (4-0)^2} \\
 &= \sqrt{25} = 5
 \end{aligned}$$

$$\begin{aligned}
 \text{dot product}(A, C) &= x_1 \cdot x_2 + y_1 \cdot y_2 \\
 &= 3 \times 2 + 4 \times 1 \\
 &= 10
 \end{aligned}$$

$$\begin{aligned}
 \text{dist}(A, C) &= \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2} \\
 &= \sqrt{(3-2)^2 + (4-1)^2} \\
 &= \sqrt{1+9} \\
 &= \sqrt{10} = 3.16
 \end{aligned}$$

$$\text{Norm}(C) = \sqrt{2^2 + 1^2} = \sqrt{5} = 2.23$$

$$\begin{aligned}
 \text{Cosine}(A, C) &= \frac{\text{dot}(A, C)}{(\text{Norm}(A) \times \text{Norm}(C))} \\
 &= \frac{10}{5 \times 2.23} \\
 &= 0.896
 \end{aligned}$$